

# CLASS D: AUDIO AMPLIFIERS OF CHOICE

As we move towards digitisation, Class D amplifiers are widely replacing Class AB in consumer electronic devices like TV sets, DVD players, laptops and mobile phones. Let's see why



TB2924FG Class-D audio power amplifier

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**I**t is difficult to make a statistical comment on the quality of audio. What feels good to hear is good quality.

Today, audio is an integral part of most consumer electronic devices. Be it your LCD TV, laptop or mobile phone, all have an audio system as a major part of product design and feature description. Some devices have inbuilt speakers with audio amplifier, while some without speakers have headphone amplifiers where you need to connect headphones. In products like

DVD players and set-top boxes, manufacturers provide a line-out where you can connect your own amplifier and speaker.

Consumers demand high-quality, multifunction audio products that are cost-competitive and feature-rich. They expect the audio to be high-fidelity and crystal-clear regardless of the format or source.

Class D audio power amplifiers have started gaining popularity where small form-factor is desired, e.g., in laptops, netbooks, mobile phones and MP3 players, as well as ultra-thin LCD and plasma displays. These are replac-

ing the traditionally used Class AB amplifiers due to their higher power efficiency. "Power efficiency is important for most of these end products that work on batteries," shares Apoorva Awasthy, business development manager, high-performance analogue division, Texas Instruments (India).

The high efficiency of Class D amplifiers reduces the losses, helps to achieve low form factor and efficient power management, and enhances the life of the product. The market for class D amplifiers is growing at the rate of above 50 per cent annually, as mentioned in a report by Bharat Book Bureau (3B).

## Moving from analogue to digital

Basically, audio signals are analogue in nature. The analogue signal in frequency range of 20 Hz to 20 kHz can be heard through the speakers. Digital audio refers to audio that gets processed digitally.

"Audio in digital form is very easy to process using various digital signal processing algorithms that improve the quality of the audio signal," says Alvin Tan, senior marketing manager, NXP Semiconductors. Digital audio is easy to store and process.

Tan adds, "Whether the source is analogue or digital, the final audio from the output of the amplifier has to be analogue in order to drive the speakers." So if you have a digitally processed audio, you need a digital-to-analogue converter before feeding the signal to the speakers. As the audio in the circuit is a low-power signal, it